



Long-Term Effects of Supplementary Aerobic Training on Muscle Hypertrophy

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Introduction

When athletes or recreationally active individuals perform structured strength or aerobic training for an extended period of time (i.e., months to years), in due course, their skeletal muscles will display a distinct phenotype reflecting the primary overload-stimulus imposed. This is manifested in its extremes by, e.g., the marathon runner displaying a slim and fatigue-resistant muscle profile, and the powerlifter or bodybuilder showing extraordinarily large muscles. From a conceptual view, it appears unlikely that the skeletal muscle is capable of achieving the extreme features displayed by the powerlifter and the marathon runner at the same time. As has been made clear in previous chapters, the question has therefore been raised whether one can perform aerobic and strength training simultaneously, without compromising the distinct muscle adaptations that would have occurred with either exercise mode in isolation.

Although the skeletal muscle adaptations outlined in the simplified example above represent classical end-point adaptations to strength and aerobic training, it should be acknowledged that phenotype changes occur along a continuum and several muscle adaptations actually show minor specificity across exercise modes, at least in the short term. For example, whole-muscle and fiber size may to some extent increase in response to aerobic training [1, 2] and fiber type transformation occurs in the same direction regardless of exercise modality [3]. Moreover, high-volume strength training may induce capillary proliferation [4] and increased oxidative enzyme activity and aerobic capacity [5]. These examples highlight that the muscle adaptive response to exercise is complex and occur along a continuum. Thus, given the limited understanding of the mechanisms dictating classical

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adaptations to aerobic and strength training, it is perhaps not surprising that the results of studies exploring muscle hypertrophy in response to concurrent training are somewhat equivocal. The goal of this chapter is to explore the long-term effects of supplementary aerobic training on muscle hypertrophy.

Concurrent Training and Muscle Hypertrophy: The Past

It should be acknowledged that so far, the majority of concurrent training studies have not included the measure of muscle hypertrophy as an outcome variable. Thus, many studies showing incompatibilities between aerobic and strength training have only focused on strength or power development. Also, in several studies, muscle strength has been compromised by additional aerobic training whereas muscle hypertrophy has been unaffected. For example, although the original work of Hickson [6] is often cited in favor for an interference effect in terms of muscle adaptations to concurrent training, that seminal study actually did not report reduced muscle growth in response to the concurrent training regime. The measurement of muscle size was made using a tape-measure to estimate thigh girth, and the increase in this measure did not differ between groups (concurrent training vs. strength training alone) at the end of the training programme. Nonetheless, ever since, numerous studies have investigated the effect of adding aerobic training to a strength training programme with assessments of muscle hypertrophy through various measurements of muscle or fiber size before and after training.

The literature provides large inconsistencies regarding the effects of concurrent training on muscle hypertrophy. Clearly, and as pointed out in several earlier review papers (e.g., [7–10]), these inconsistencies are probably due to differences in study design factors such as study duration, subject population, age and training status, as well as differences in training mode, frequency, intensity, and volume. Indeed, a previous meta-analysis indicated that although there was no statistical difference in the averaged effect size for lower-body muscle hypertrophy between concurrent training and strength training [10], there were some differences when the effect sizes were split for different training variables. More specifically, interference in muscle hypertrophic development was noted when running was selected as the aerobic training modality, or when high training frequencies or long aerobic training workouts were employed. Since then, however, several new studies have been published including review papers challenging the idea of an interference effect on muscle hypertrophy with concurrent training [11].

Concurrent Training and Muscle Hypertrophy: Updating the View

A summary of 23 published studies [3, 6, 12–33] reporting changes in muscle hypertrophy to concurrent training and strength training alone are summarized in Table 12.1. The subject populations in the studies identified have typically been untrained or recreationally active young men, with a few exceptions where studies

Table 12.1 Summary of concurrent training studies measuring muscle hypertrophy compared with strength training alone

Study	Subjects	Training duration and frequency	Order of sessions	Training programme	Hypertrophy measure	Interference (yes/no)	Comment
Hickson [6]	Recreationally trained men and women	5 days/week, 10 weeks	Unclear order. ~2 h between sessions	ST: 3–5 sets × 5 reps in isoinertial leg exercises, 80% of 1RM. AT: Cycling intervals and continuous running	Thigh girth	N	
Nelson et al. [16]	Untrained men	4 days/week, 20 weeks	ST—AT	ST: 3 sets × 6 reps max knee extensions @ 30 deg/s. AT: 30–60 min continuous cycling	Fiber area	N	Greater increase in AT + ST than ST
Sale et al. [17, 36]	Untrained men and women. Between leg comparison	3 days/week, 22 weeks	AT—ST	ST: 6 × 15–20 reps leg press. AT: 5 × 3 min interval cycling at 90–100% of VO2max	CSA (CT), thigh girth, and fiber area	N	
Craig et al. [18]	Untrained men	3 days/week, 10 weeks	AT—ST	ST: 3 × 6–10 reps of UB and LB exercises including LP and leg curls. AT: continuous running 30–35 min @ 75% MHR	Thigh girth	Y	No interaction reported. Very small increase overall
Volpe et al. [19]	Untrained women	3 days/week, 9 weeks	AT—ST	ST: 2–4 sets of 4–10 reps of LP, extensions, leg curl, calf raise + 6 UB exercises. AT: 20–25 min of running at 75% of predicted MHR	Thigh girth	N	
Kraemer et al. [3]	Trained (army) men	4 days/week, 12 weeks	AT—5–6 h—ST	ST: 3–5 sets of 5–15 reps, 10 UB exercises and 8 isoinertial leg exercises. AT: distance and interval running with aim to maximally stress the aerobic system	Fiber area	Y	Interference in Type 1 and 2C fiber area (no interaction reported)

(continued)

Table 12.1 (continued)

Study	Subjects	Training duration and frequency	Order of sessions	Training programme	Hypertrophy measure	Interference (yes/no)	Comment
McCarthy et al. [20]	Untrained men	3 days/week, 10 weeks	Same session, rotating order	ST: 4 × 5–7 reps, isoinertial strength such as squats, BP, extensions, leg curls, and UB exercises. AT: 50 min continuous cycling @ 70% HR reserve	Thigh girth	N	
Bell et al. [27]/Putman et al. [28]	Recreationally trained men	3 days/week, ST + AT 6 days/week, 12 weeks	Alternate days	UB exercises free weights + machines. AT: Cycling, continuous + intervals	Fiber area	Y	Interference in Type 1 fiber area (no interaction reported)
McCarthy et al. [21]	Untrained men	3 days/week, 10 weeks	Same session, rotating order	ST: 3 sets × 5–7 reps of 8 exercises including squats, curls, extensions. AT: continuous cycling	CSA (CT) and fiber area	N	
Häkkinen et al. [22]	Recreationally trained men	2 days/week, ST + AT 4 days/week, 21 weeks	Alternate days	ST: 3–5 sets of 5–15 reps of LP and KE + 4 UB exercises. Aim both maximal strength and explosive strength. AT: First 30 min cycling or walking, then increased cycling at different intensities for 60–90 min	CSA (MRI)	N	
Izquierdo et al. [23]	Untrained older men	2 days/week, 16 weeks	Alternate days	ST: first 8 weeks: 3–4 sets of 10–15 reps @ 50–70% 1RM, last 8 weeks: 3–5 sets × 5–6 reps of LP, KE, BP and 4–5 other UB exercises. AT: Continuous and interval cycling	CSA (ultrasound)	N	
Izquierdo et al. [12]	Untrained middle-aged men	2 days/week, 16 weeks	Alternate days	ST: first 8 weeks: 3–4 sets of 10–15 reps @ 50–70% 1RM, last 8 weeks: 3–5 sets × 5–6 reps of LP, KE, BP and 4–5 other UB exercises. AT: Continuous and interval cycling	CSA (ultrasound)	N	

Sillanpää et al. [13]	Untrained middle-aged men	2 days/week, ST + AT 4 days/week, 21 weeks	Alternate days	ST: whole-body programme including 2 exercises for the leg extensors. AT: 30–60 min continuous cycling	VL + VI thickness (ultrasound)	N	
Ahtiainen et al. [24]	Untrained older men	2 days/week, ST + AT 4 days/week, 21 weeks	Alternate days	ST: whole-body programme including 2 exercises for the leg extensors. AT: 30–60 min continuous cycling	VL thickness and fiber area	N/Y	Interference in Type 2 fiber area (no interaction reported) but not in VL thickness
Karavirta et al. [14]	Untrained older men	2 days/week, ST + AT 4 days/week, 21 weeks	Alternate days	ST: whole-body programme including 2 exercises for the leg extensors. AT: 30–60 min continuous cycling	Fiber area	Y	Interference in 2 fiber area (no interaction reported)
Mikkola et al. [15]	Untrained middle-aged men	2 days/week, ST + AT 4 days/week, 21 weeks	Alternate days	ST: whole-body programme including 2 exercises for the leg extensors. AT: 30–60 min continuous cycling	CSA (MRI)	N	Greater increase in AT + ST than ST
de Souza et al. [25, 26]	Untrained young men	2 days/week, 8 weeks	Same session, rotating order	ST: 3–5 sets of 6–12 RM LP, KE, Knee flexion. AT: Interval training at 80–100% of vVO2max	CSA (MRI) and fiber area	N/Y	Interference in fiber area (no interaction reported) but not whole-muscle MRI
Jones et al. [29]	Recreationally resistance-trained men	3 days/week, 6 weeks	ST—AT	ST: 5 sets of 6 reps at 80% of MVC in isokinetic leg extensions. Unilateral training. AT: 30 min of repeated extensions at 30% of MVC	Thigh girth	Y/N	Interference with 3 days/week but not with 1 day/week AT
Lundberg et al. [30]	Moderately trained men. Between leg comparison	AT: 3 days/week, ST: 2 or 3 days/week, 5 weeks	AT—6 h—ST	ST: 4 × 7 sets of flywheel knee extensions. AT: one-legged continuous cycling ~45 min	Volume and CSA (MRI), fiber area	N	Greater increase in AT + ST than ST

(continued)

Table 12.1 (continued)

Study	Subjects	Training duration and frequency	Order of sessions	Training programme	Hypertrophy measure	Interference (yes/no)	Comment
Lundberg et al. [31]	Moderately trained men. Between leg comparison	AT: 3 days/week, ST: 2 or 3 days/week, 5 weeks	AT—15 min—ST	AT: one-legged continuous cycling ~45 min. ST: 4 × 7 sets of flywheel knee extensions	Volume and CSA (MRI)	N	Greater increase in AT + ST than ST
Kazior et al. [32]	Untrained young men	2–4 days/week (progressive increase), 7 weeks	AT—ST	AT: Both continuous and interval cycling sessions. ST: Leg press 4–6 sets × 8–15 reps	Fiber area	N	Greater increase in AT + ST than ST
Terzis et al. [33]	Moderately trained young women	3 days/week, 6 weeks	ST—AT	AT: 30 min low-intensity running 60–70% of MHR. ST: Lower-body power training (ECC half-squats, CMJ jumps, drop-jumps)	Fiber area	Y	Interference in all fiber types
Tsitkanou et al. [34]	Recreationally active young men	2 days/week, 8 weeks	ST—AT	AT: High-intensity cycling 10 sets of 1 min @maximal aerobic power. ST: Leg press and half-squats 4 sets of 6RM	CSA (ultrasound) and fiber area	N	

Note: ST strength training, AT aerobic training, RM repetition maximum, LP leg press, KE knee extension, UB upper body, MHR maximal heart rate, LB lower body, BP bench press, CSA cross-sectional area, CMJ counter movement jump, VO₂ oxygen uptake, ECC eccentric, MVC maximal voluntary contraction, CT computerized tomography, MRI magnetic resonance imaging, VL vastus lateralis, VI vastus intermedius

have looked at women or trained/athletic populations. The study duration has varied between 5 and 22 weeks, and the training frequencies have varied between 2 and 6 days per week (most often with 2 or 3 weekly strength and/or endurance training sessions). In 15 of the studies, the strength and aerobic sessions have been carried out on the same day, typically with only a short rest period in between the sessions. In seven studies, the sessions were scheduled on alternative days.

There Is No Uniform Interference Effect on Muscle Hypertrophy

When summarizing the effect sizes and the relative increase in muscle hypertrophy across all studies, it becomes apparent that there is no general or “global” interference effect on muscle hypertrophy from concurrent training (Fig. 12.1). The global average increase in muscle hypertrophy (i.e., averaging all measurements from all studies) is 14% from strength training alone and 12% with concurrent aerobic and strength training ($P = 0.36$). The average effect size for muscle hypertrophy is 0.76 for strength exercise alone and 0.60 for concurrent exercise ($P = 0.17$). If each study is given the same weight in the analysis (regardless of whether they included several measures of hypertrophy or not), the results are still very similar. Indeed, the majority of the individual studies independently report no difference in the mean hypertrophic response across the two training conditions (Table 12.1, Fig. 12.1). It should be noted that there are as many papers reporting a greater increase in muscle hypertrophy with concurrent training as there are papers showing an interference effect.

An important research design note is that in those studies reporting some kind of interference effect, all except one study used fiber size as the outcome measure of muscle hypertrophy. As the only exception, Craig et al. [18] used the thigh girth to assess muscle hypertrophy pre- and post-training, and the authors noted an increase of 1.8% in the strength group compared with 1.3% increase in the concurrent training group. Even if the author concluded that the increase in the strength group was significant in contrast to the increase in the concurrent group, one must certainly question if this small difference in thigh girth can be meaningful. An important point to make here is that none of the studies showing an interference effect used the most reliable techniques to assess muscle size (i.e., MRI or CT). In some cases, however, these techniques have been used together with fiber size measurements [17, 21, 25, 26, 30]. However, in the study by de Souza et al., the interference effect noted in fiber size measurements were not noted at the whole-muscle level using MRI or CT [25, 26]. It is difficult to elucidate the reason for these conflicting findings but two main hypotheses can be postulated. Since MRI and CT measure whole-muscle anatomical cross-sectional area, it is possible that changes in fiber size accompanied by changes in muscle architecture (i.e., increased pennation angle) will mask early hypertrophic effects that are not detected by either of these techniques. Thus, it could be speculated that with longer training periods, the interference effect would be noted also at the whole-muscle level with CT/MRI imaging. Alternatively, and perhaps even more likely, the conflicting findings on fiber size differences could reflect random effects or

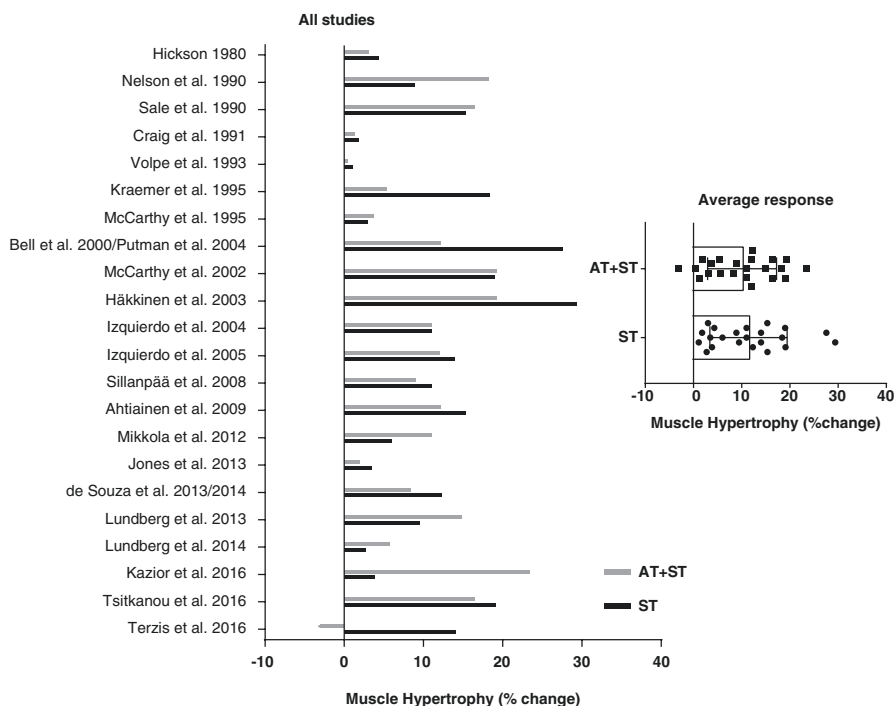


Fig. 12.1 A quantitative summary of the relative increase in muscle hypertrophy in all studies included in Table 12.1. The panel to the left depicts each individual study whereas the right panel shows the average response. *ST* strength training alone, *AT + ST* concurrent aerobic and strength training

type 1 errors that occur due to low sample sizes and a lack of a complete data sets, i.e., all fibers of the entire muscle, which leads to the extreme variance that is associated with measuring fiber size from a small sample of muscle tissue. To complicate this issue further, the statistics used to reach the conclusions in Table 12.1 can sometimes be questioned. Specifically, when one compares two groups across two time points in order to assess if the experimental condition (concurrent training) respond differently than the control condition (strength training), the main outcome statistics should be the interaction effect between group and time. However, in many of the studies shown in Table 12.1, and actually in all of the studies claiming an interference effect, the authors instead have analyzed the effects of each training mode (strength training vs. concurrent training) separately, obviously providing a significant bias.

The Effect of Different Training Variables on the Interference Effect in Muscle Hypertrophy

Training Frequency

Given that there is no general interference effect of concurrent training on hypertrophy, it appears clear that specific programme variables such as training

frequency, volume, and how the sessions are distributed will dictate the outcome of a specific concurrent training regime. Only one study has directly assessed the influence of endurance training frequency on the hypertrophic response to concurrent training. In that study [29], subjects were randomized to an experimental condition of either (a) strength training, (b) concurrent training with the aerobic training carried out 3 days/week, or (c) concurrent training with the aerobic training carried out 1 day/week. The strength training was always performed 3 days/week and the study duration was 6 weeks. The results showed a similar hypertrophic response (limb girth) between strength training alone and concurrent training with low-frequency aerobic training. Interestingly, however, the higher frequency of aerobic training was associated with an interference effect. While these data are the only direct evidence supporting that greater frequencies of aerobic training are negative for the muscle hypertrophic response to strength training, the data summarized in the current chapter also lend slight support to this notion. More specifically, if one divides all the studies in Table 12.1 into two groups split on training frequencies of either >3 days/week or ≤ 3 days/week, there seems to be a tendency for an interference effect with concurrent training with high training frequencies as compared with low frequencies (Fig. 12.2a).

Aerobic Training Modality

Regarding the effect of exercise mode, only four studies in Table 12.1 have selected running as the aerobic exercise modality [3, 6, 18, 33], whereas the remaining studies have studied cycling. Even though there is not enough data to draw firm statistical conclusions on any differential effect on muscle hypertrophy depending on the selected aerobic training mode, it is worth noting that three of the four studies employing running as the aerobic mode actually do show an interference effect (Fig. 12.2b). This strongly suggests that running, as compared with cycling, will exacerbate the risk of an interference effect on muscle hypertrophy. In fact, this may not be surprising since cycling in itself has often, in contrast to running [35], been associated with muscle hypertrophy [2].

Training Distribution

Regarding the distribution of training sessions, i.e., whether it is better to schedule the different training sessions on the same or alternate days, there is no difference in the outcome of muscle hypertrophy between concurrent training and strength training (Fig. 12.2c). This is also supported by Sale et al. [36] reporting that same day (vs. different day) concurrent training did not impede muscle hypertrophy (even though increases in muscle strength were attenuated by same day training). However, it is worth noting that there is a general main effect indicating that overall, training on alternate days gives better end-point results (hypertrophy) for both

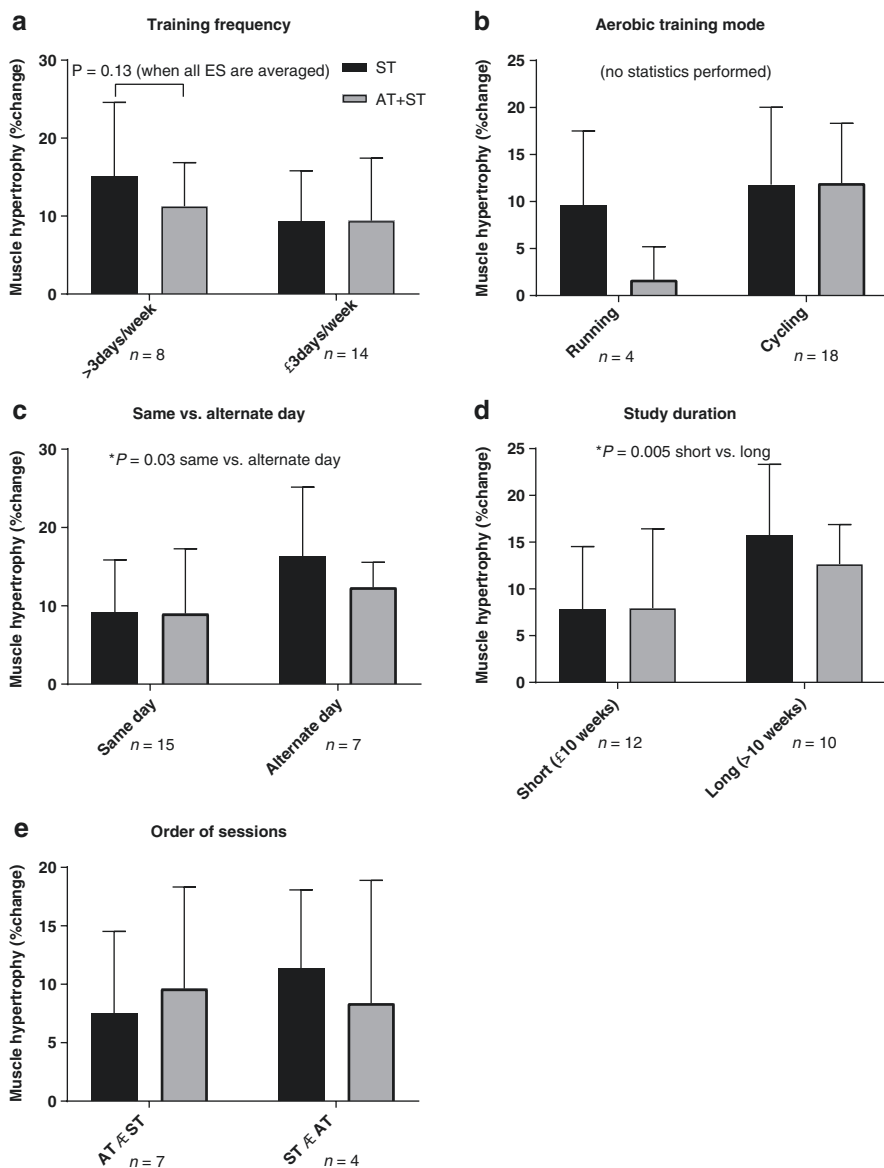


Fig. 12.2 The effect of different training variables on the relative increase in muscle hypertrophy. The indicated sample sizes (n) reflects the number of individual studies included in the analysis. ST strength training alone, AT + ST concurrent aerobic and strength training

strength training and concurrent training (Fig. 12.2c). Similarly, regarding the effect of study duration, longer study durations (>10 weeks) generally produces greater hypertrophy than shorter studies (Fig. 12.2d), but there is no difference between strength training and concurrent training in this regard.

Order of Sessions

There is somewhat limited data to quantify the effect of different orders of the session (i.e., aerobic training before or after strength), but from the 11 studies that have scheduled the aerobic and strength sessions in a specified order on the same day, there seems to be no clear order effect (Fig. 12.2e). In addition to these studies, some investigators have assessed the order effect by comparing two concurrent training groups with opposite order of the exercise modes with no comparison to strength training alone. The collective interpretation from these studies is also that there is no clear order effect on muscle hypertrophy [37, 38]. This notion certainly questions the often purported view that performing aerobic training prior to resistance training would activate molecular signalling pathways that are antagonistic to those induced by subsequent strength training, which in turn would inhibit muscle hypertrophy [8, 39]. Regardless of the order of training sessions, an interesting finding reported by K  ismaa et al. was that concurrent training in the evening led to larger gains in muscle mass compared with the same training programme in the morning hours [38], suggesting that the timing of the concurrent training session is more important than the order of the different exercise modalities.

Other Factors

Unfortunately, there is not enough data to reliably quantify any effect of trained vs. untrained populations, women vs. men, different intensities during the aerobic training, different length of recovery between sessions, or upper- vs. lower-body muscle groups. There are, however, some preliminary data from individual studies that are worth mentioning. In my laboratory, we carried out two 5-week training studies where the only training variable manipulated between the studies was the length of recovery between the aerobic and strength training carried out on the same day 2–3 days per week. Regardless of whether the recovery was 6 h or 15 min between sessions, muscle hypertrophy was substantially greater following the concurrent training regime in recreationally active young men [30, 31]. Although this indicates that for muscle growth it does not matter if daily sessions are performed combined or with several hours of rest in between, it should be acknowledged that some aspects of strength performance were negatively influenced by the shorter recovery period.

Another study, performed by R  nnestad et al., explored the effects of high-volume aerobic training on the muscle hypertrophic response to resistance training in well-trained cyclists [40]. These cyclists, with limited strength training experience, performed heavy strength training twice a week in addition to high volume (~10 h per week) aerobic training during a 12-week period. A group of non-strength trained individuals performed the same strength training as the cyclists, but without added aerobic (cycling) training. Following the training period, the relative increase in thigh cross-sectional area was greater in the strength training group than in the concurrent training group, suggesting that high volumes of aerobic training impairs

muscle hypertrophy in trained individuals. These results should, however, be viewed with some caution since the strength training group consisted of non-cyclists. This was also why that particular study did not qualify for inclusion in the quantitative summary in this chapter.

Regarding the effect of aerobic training intensity, Fyfe et al. determined the effects of concurrent training incorporating either high-intensity interval training or volume-matched moderate-intensity continuous training [41]. The authors noted that lower-body lean mass increased similarly between the groups of strength training alone and moderate-intensity training + strength training, whilst the hypertrophic response was attenuated in the group incorporating high-intensity aerobic training with the strength training. Although these data indicate that high-intensity may be worse than low-intensity aerobic training for the development of muscle hypertrophy, the effect sizes were rather small and the responses were not reflected in the training-induced changes in strength. Therefore, it remains for future studies to confirm or refute these preliminary observations.

Summary

The collective body of evidence indicates that no universal interference effect of aerobic training on muscle hypertrophy seems to exist when added to a strength training programme. Thus, the effects of concurrent training on muscle hypertrophy ultimately depend on the specific training programme carried out. High training frequencies (>3 days/week regardless of training mode) and the running mode of exercise appear to increase the risk of a compromised hypertrophic response to strength training compared with low training frequencies and the cycling modality. Unfortunately, however, there is a lack of systematic high-powered studies examining the effect of different training variables on the concurrent training effect. Nonetheless, current data suggest that most individuals can accomplish significant muscle hypertrophy with strength training despite concurrently performing aerobic training. From a practical standpoint, it could be recommended that athletes or trained individuals prioritizing gains in muscle mass and strength should choose cycling instead of running as the aerobic modality and avoid high aerobic training frequencies. It remains for future studies to determine for how long an athlete or recreationally active individual can maintain uncompromised hypertrophic development before any (potential) interference effect becomes evident.

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